

Chronometry from Crossed Null Phases

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral;
see RECOVERY_PROVENANCE.md.

5. Exact crossed-phase theorem

Let ϕ_+ and ϕ_- be circle-valued phases with null gradients

$$k_a = \nabla_a \phi_+, \quad \ell_a = \nabla_a \phi_-,$$

$$g^{ab} k_a k_b = g^{ab} \ell_a \ell_b = 0.$$

Assume they are nonparallel and define

$$C = -\frac{1}{2} g^{ab} k_a \ell_b > 0.$$

Set

$$T = \frac{\phi_+ + \phi_-}{2}, \quad R = \frac{\phi_+ - \phi_-}{2},$$

and introduce the conformally invariant tensor

$$h_{ab} = C g_{ab}.$$

Because

$$\nabla T = \frac{k + \ell}{2}, \quad \nabla R = \frac{k - \ell}{2},$$

the inverse Gram matrix under g is

$$\begin{pmatrix} \nabla T \cdot \nabla T & \nabla T \cdot \nabla R \\ \nabla R \cdot \nabla T & \nabla R \cdot \nabla R \end{pmatrix} = \begin{pmatrix} -C & 0 \\ 0 & C \end{pmatrix}.$$

Since $h^{ab} = C^{-1} g^{ab}$,

$$\boxed{\begin{aligned} h^{ab} \nabla_a T \nabla_b T &= -1, \\ h^{ab} \nabla_a R \nabla_b R &= +1, \\ h^{ab} \nabla_a T \nabla_b R &= 0. \end{aligned}}$$

Thus two crossed null phases define a unit clock direction and a unit ruler direction in their two-plane. Under $g_{ab} \rightarrow \Omega^2 g_{ab}$,

$$C \rightarrow \Omega^{-2}C,$$

so $h_{ab} = Cg_{ab}$ is invariant.

In flat spacetime, choose

$$\phi_+ = \omega_+(t - x), \quad \phi_- = \omega_-(t + x).$$

Then

$$C = \omega_+\omega_-, \quad \eta = \frac{1}{2} \ln \frac{\omega_+}{\omega_-}.$$

Therefore

$$\boxed{\omega_+\omega_- = \text{clock or mass scale}^2, \quad \frac{\omega_+}{\omega_-} = \text{rapidity data.}}$$

The wave sum factorises as

$$e^{i\phi_+} + e^{i\phi_-} = 2e^{iT} \cos R.$$

The mean phase is a temporal carrier and the difference phase supplies standing-wave nodes. This is the algebraic skeleton of a clock-and-ruler pair.

Novelty correction

The exact map is not a new conformal completion. Writing $\phi_{\pm} = T \pm R$ and imposing exact nullity gives

$$C = -(\nabla T)^2,$$

and hence

$$h_{\mu\nu} = -(g^{\alpha\beta} \partial_{\alpha} T \partial_{\beta} T) g_{\mu\nu},$$

which is the singular conformal map of mimetic gravity, with R supplying an orthogonal equal-norm ruler field [Mimetic2013]. The theorem remains exact and useful; the broad novelty claim does not.

6. Homogeneous locking potential and its limitation

The first proposed mechanism introduced a mass-squared order parameter Σ and

$$V(C, \Sigma) = \frac{\kappa}{2}(C - \Sigma)^2 + \frac{\beta}{4}\Sigma^2 \left(\ln \frac{\Sigma}{\Lambda^2} - \frac{1}{2} \right) + \frac{\beta}{8}\Lambda^4.$$

The stationary point

$$C = \Sigma = \Lambda^2$$

has Hessian

$$H = \begin{pmatrix} \kappa & -\kappa \\ -\kappa & \kappa + \beta/2 \end{pmatrix},$$

so

$$H_{11} = \kappa > 0, \quad \det H = \frac{\kappa\beta}{2} > 0.$$

For $\kappa, \beta > 0$, this is a strict homogeneous minimum. It demonstrates a possible algebraic lock of a crossed phase invariant to a transmuted scale. It does **not** establish kinetic rank, hyperbolicity, absence of ghosts, or a healthy $3 + 1$ dimensional field theory.

That distinction proved decisive.